

BEYOND GENERAL RELATIVITY

for
babies
students

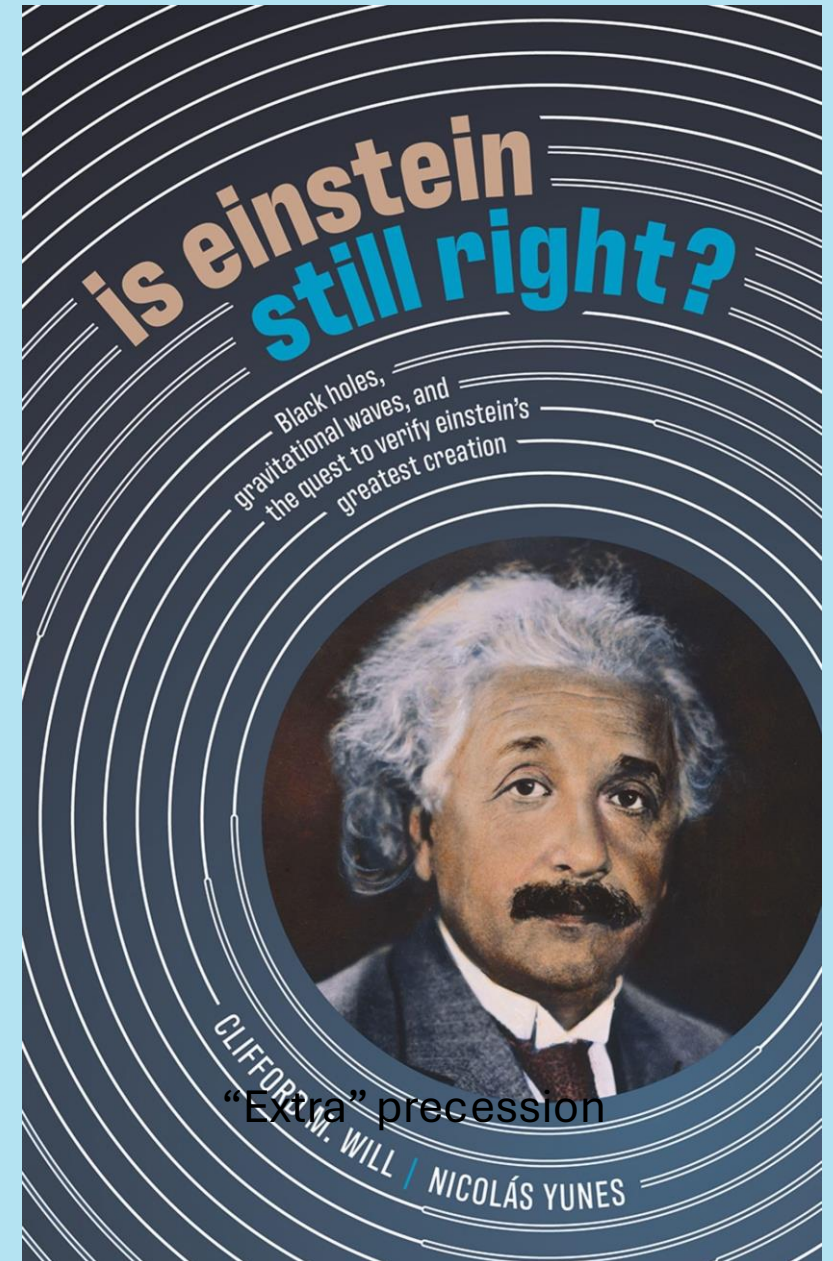


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vipser summer
school

July 27, 2026

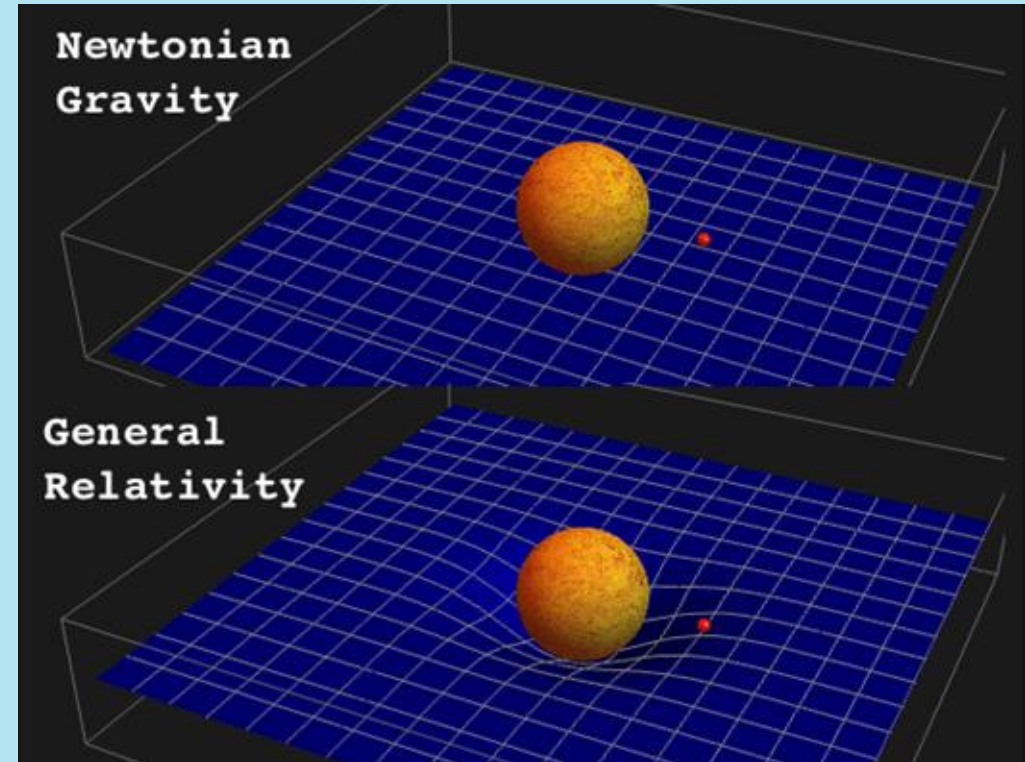
GR works! So far...

- GR has passed every test to date.



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 - the precession of the perihelion of Mercury



“Extra” precession

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- it can explain...
 - the precession of the perihelion of Mercury
 - the amount by which light is redshifted in a gravitational field

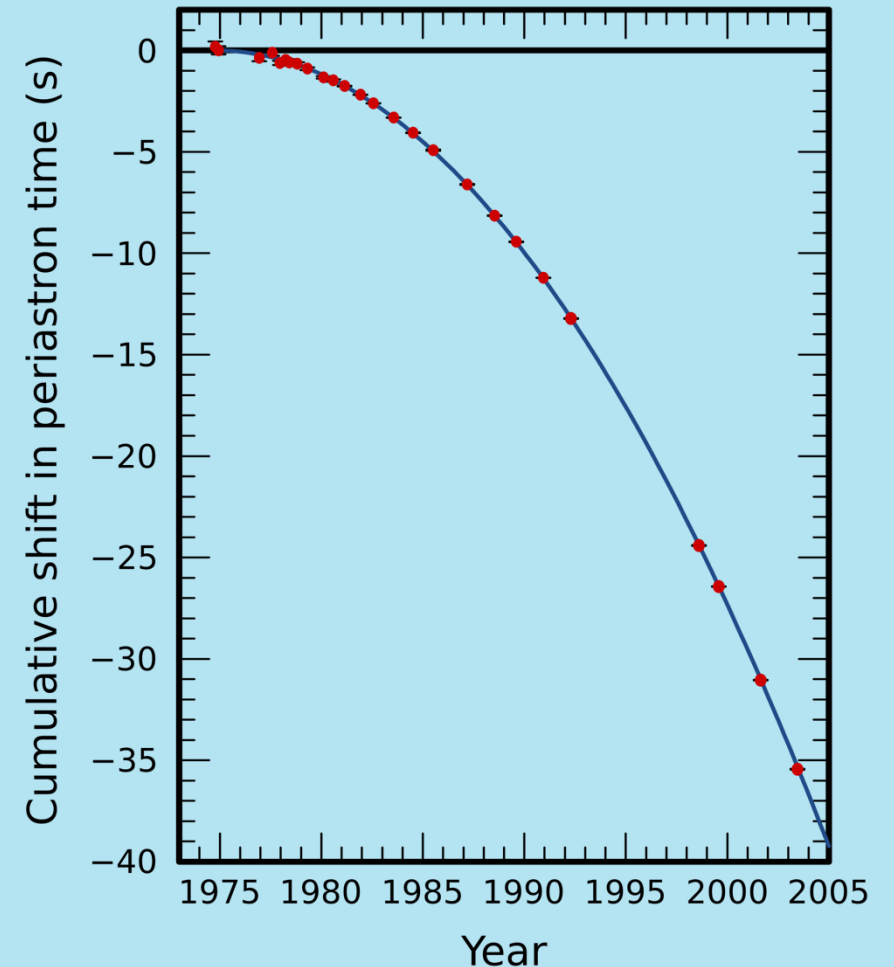
Image credit: Bettmann/CORBIS



GR works! So far...

- GR has passed every test to date.
- it can explain...
 - the precession of the perihelion of Mercury
 - the amount by which light is redshifted in a gravitational field
 - the shrinking of the orbit of a binary pulsar system – Hulse-Taylor pulsar

[Weisberg, Taylor, 2004]



Why continue to test GR?

- the goal of science is to predict and verify
- do this across multiple regimes
- weak field regime – e.g., solar system observations

$$\nu \ll 1, \quad \mathcal{C} \ll 1 \Rightarrow \mathcal{M} \ll \mathcal{R}$$

- strong field regime – e.g., binary pulsars

$$\nu \ll 1, \quad \mathcal{C} \lesssim \mathcal{O}(1) \Rightarrow \mathcal{M} \lesssim \mathcal{R}$$

- extreme gravity regime – e.g., compact binary coalescences

$$\nu \lesssim \mathcal{O}(1), \quad \mathcal{C} \lesssim \mathcal{O}(1) \Rightarrow \mathcal{M} \lesssim \mathcal{R}$$

$$\mathcal{C} = \frac{\mathcal{M}}{\mathcal{R}}, \quad \nu$$

Lots of tests in
70s, 80s

Start in 70s

New
opportunities!

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Why propose theories “beyond GR”?

- While GR is a beautiful theory, we could be missing something...
- Example - no fully consistent theory of quantum gravity
- There are all kinds of different modified theories proposed to answer different open questions in physics

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To test or modify GR, we need to understand
why it is special...

The action in general relativity

- The Einstein-Hilbert action:

$$S = \int d^4x \sqrt{-g} R$$

where $R = g^{\mu\nu} R_{\mu\nu}$

From action → field equations

- Remember from a classical mechanics course:

$$S = \int dt L(q, \dot{q}) \quad \delta S = 0 \Rightarrow \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0$$

- In GR we vary the action with respect to the metric

$$S = \int d^4x \sqrt{-g} R$$

$$\delta S = \int d^4x \left[g^{\mu\nu} R_{\mu\nu} \delta \sqrt{-g} + \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} + \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} \right]$$

See notes on [Github](#) and/or [Carroll Ch. 4](#)

$$\Rightarrow G_{\mu\nu} = 8\pi T_{\mu\nu}$$

From Lovelock's theorem:

In four spacetime dimensions, the only symmetric, divergence-free rank-2 tensor constructible locally from the metric and its derivatives up to second order is

$$E_{\mu\nu} = aG_{\mu\nu} + b g_{\mu\nu}$$

where a and b are constants and $G_{\mu\nu}$ is the Einstein tensor.

GR is unique

General relativity is **the** unique metric theory of gravity **in 4D** that satisfies:

1. The metric is the only dynamical field (no torsion or additional fields)
2. The eqns of motion contain at most second derivatives of g

Road map:

$$g_{\mu\nu} \rightarrow \Gamma(g, \partial g) \rightarrow R^\alpha{}_{\beta\gamma\delta}(\Gamma, \partial\Gamma) \rightarrow \underbrace{R_{\alpha\beta}(R^\alpha{}_{\beta\gamma\delta})} \rightarrow R(R^\alpha{}_\alpha)$$

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \rightarrow G_{\mu\nu} = \delta\pi T_{\mu\nu}$$



GR is unique

General relativity is **the** unique metric theory of gravity **in 4D** that satisfies:

1. The metric is the only dynamical field (no torsion or additional fields)
2. The eqns of motion contain at most second derivatives of g
3. Field eqns depend only on the metric and its derivatives at a single point (local)
4. Theory is diffeomorphism invariant (holds in all coordinate systems)

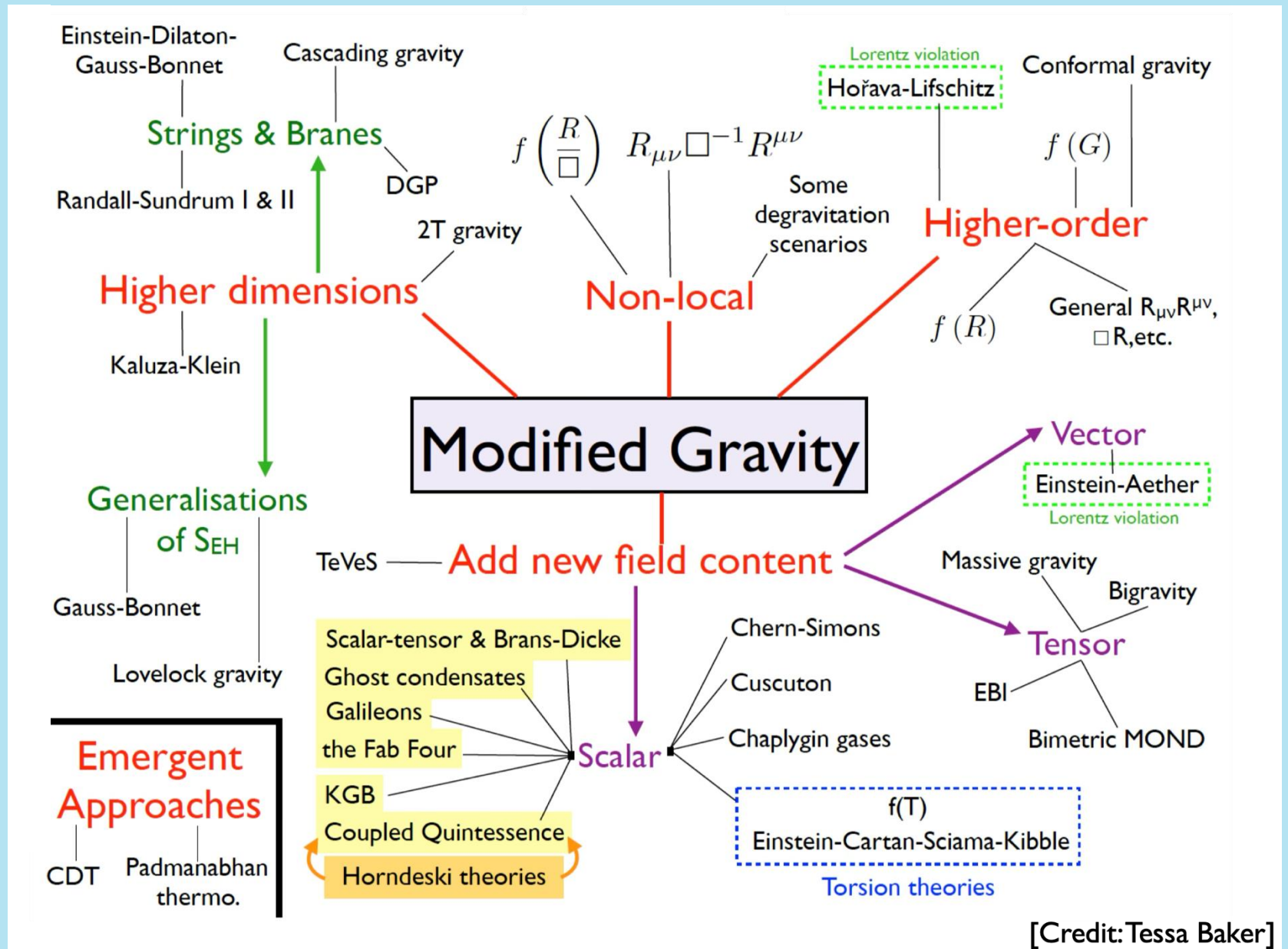
How to modify/test GR

Intentionally break the assumptions of GR

1. The metric is the only dynamical field (no torsion or additional fields)
→ Consider other fields
2. The eqns of motion contain at most second derivatives of g
→ Allow higher than 2nd derivatives of the metric in field eqns
3. Field eqns depend only on the metric and its derivatives at a single point (local)
→ Give up locality
4. In 4D → Work in a higher dimensional space

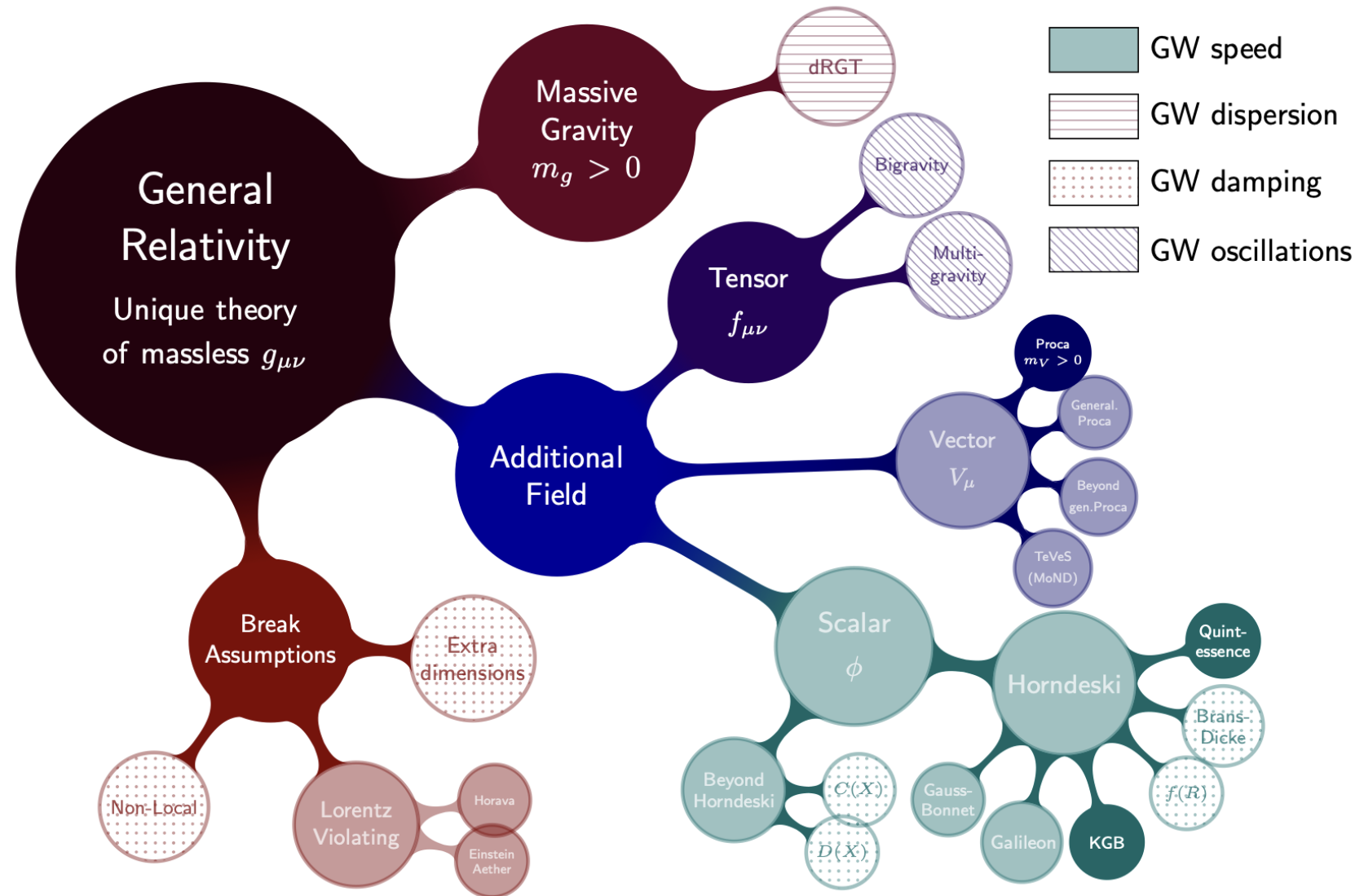
Some theories will break more than one at a time, but this is a useful way to think about modifying GR.

A “zoo” of modified theories



[Credit: Tessa Baker]

Modified gravity roadmap



[Ezquiaga & Zumalacárregui, 2018]

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Infinitely many possibilities...

- Can always add something else/more terms to the action
- BUT any new theory had better satisfy every test to date (which is hard to do)
- With so many possibilities, how do we even decide what to test??

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Theory specific vs theory agnostic

- Theory specific tests can be more sensitive – we can include more assumptions/physical information
 - Sometimes we have new opportunities to test/constrain a specific theory
- Theory agnostic tests allow us to constrain more than one theory at a time. A few examples:
 - ppE framework
 - search for a unique signature of modified theories (like additional polarizations or modified propagation speed)

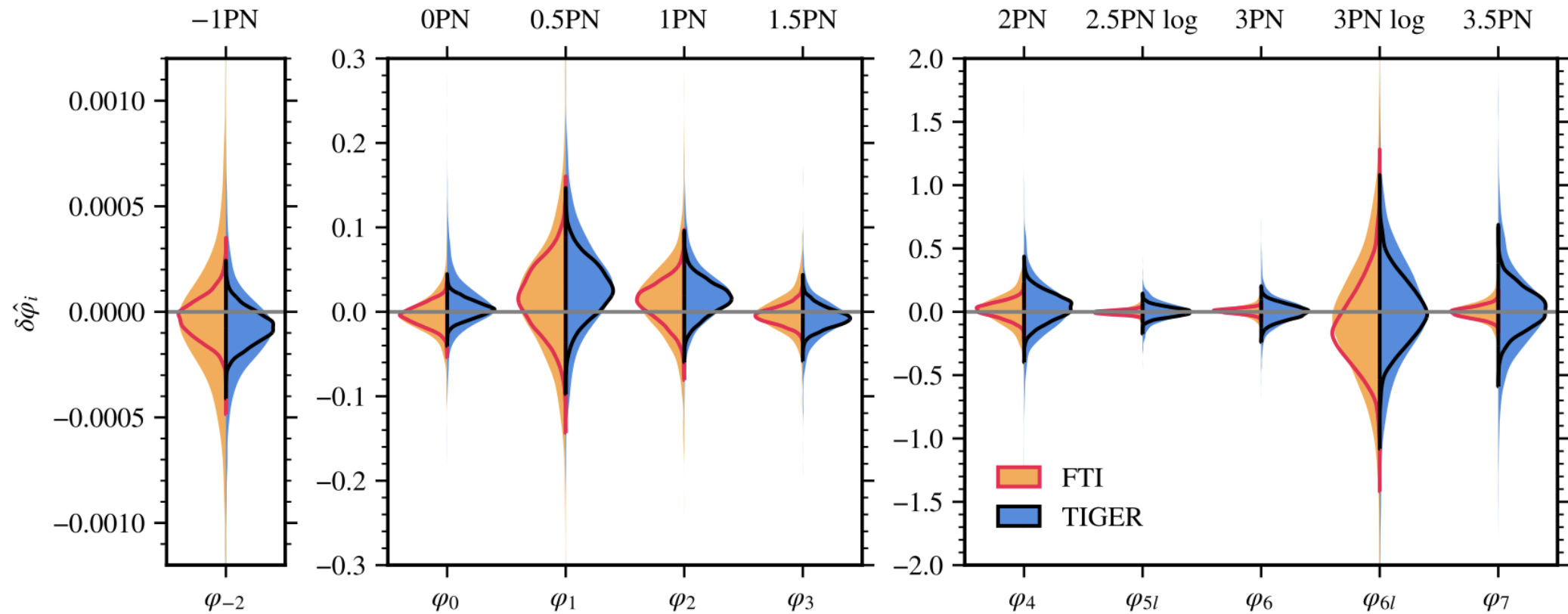
ppE framework

- ppE = parameterized post-Einsteinian
- The basic idea: modifications to GR are small. Let's characterize these modifications by adding a small term to both the amplitude and phase of the wave

$$\tilde{h} \sim (1 + \alpha_{\text{ppE}} u_2^a) h_{\text{GR}} e^{i\beta_{\text{ppE}} u_2^b}$$

- We can search for deviations order by order – this is what's done in the LVK test of GR papers

Example: test of phase deviation in GWTC-5 (LVK)



Additional polarizations?

Detecting an additional polarization would be definitive evidence of new physics.

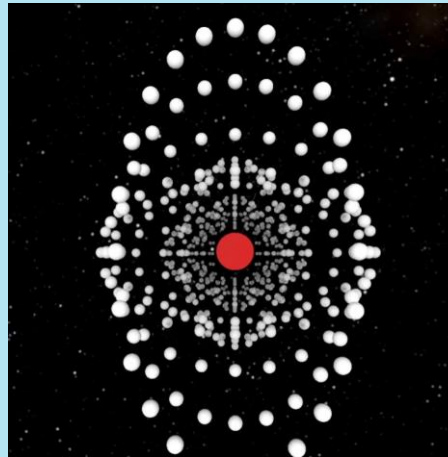
Gravitational waves in GR

- Two tensor polarizations
- Propagate with speed $v_T = c$

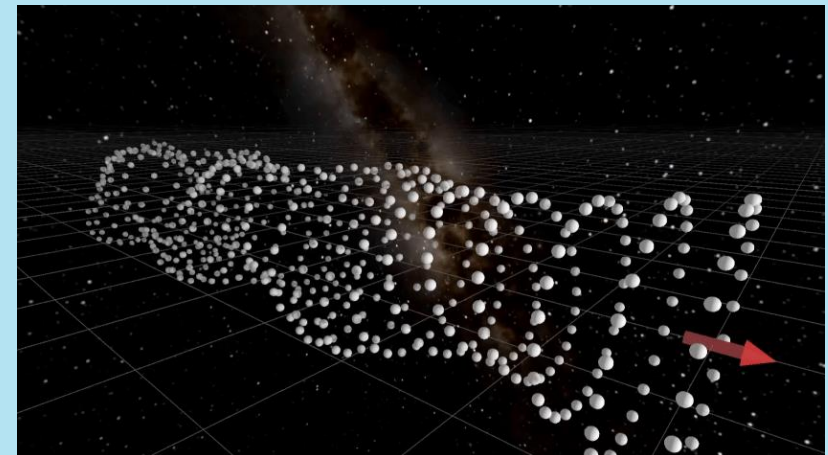
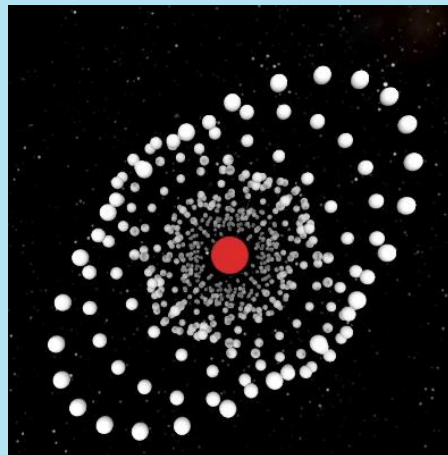
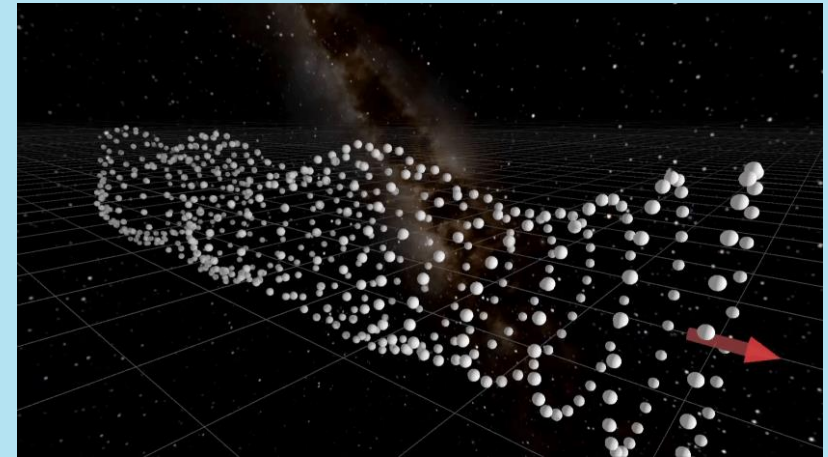
Animation credit:
[\[Schumacher et al. ISVC, 2024; POINT\]](#)



GW out of screen



45° view, GW along direction of arrow



Additional vector polarizations

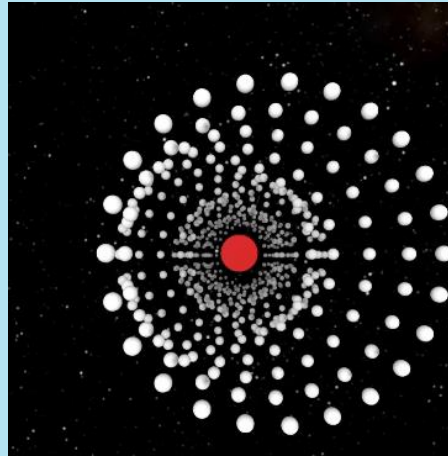
- Two vector polarizations
- Propagate with speed v_V

Animation credit:
[\[Schumacher et al. ISVC, 2024; POINT\]](#)

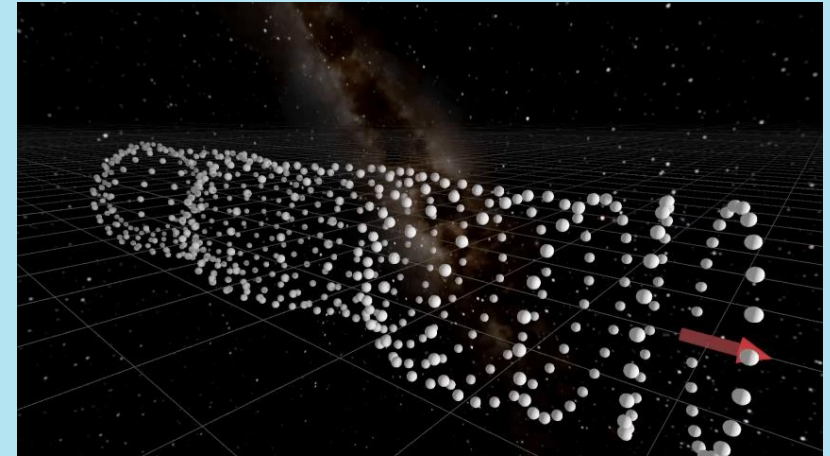


X

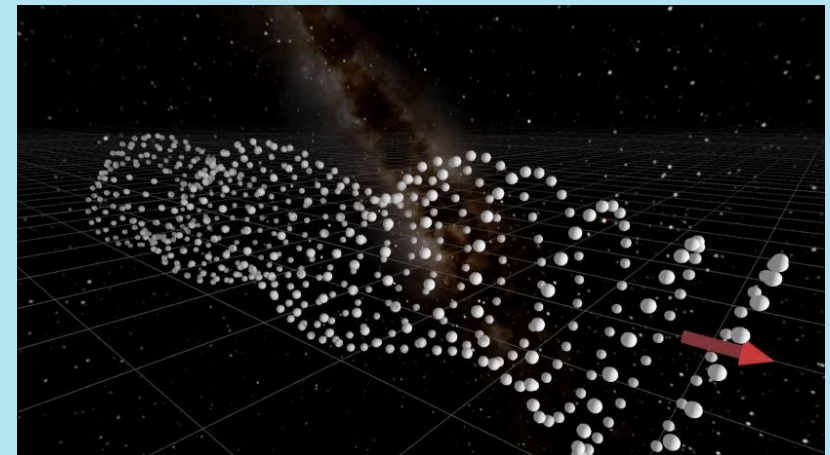
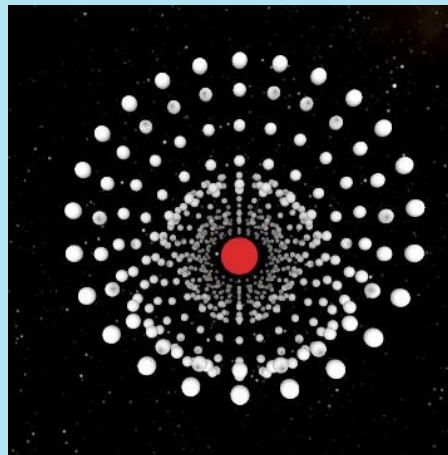
GW out of screen



45° view, GW along direction of arrow



Y



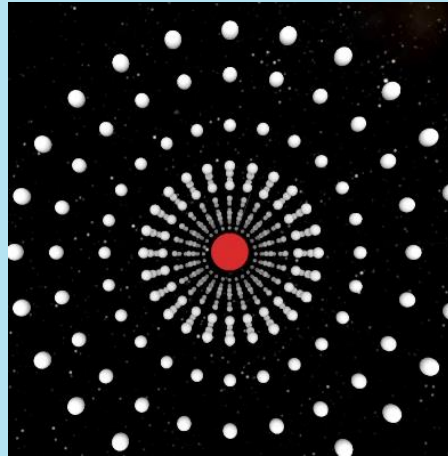
Additional scalar polarizations

- Two scalar polarizations
- Propagate with speed v_s

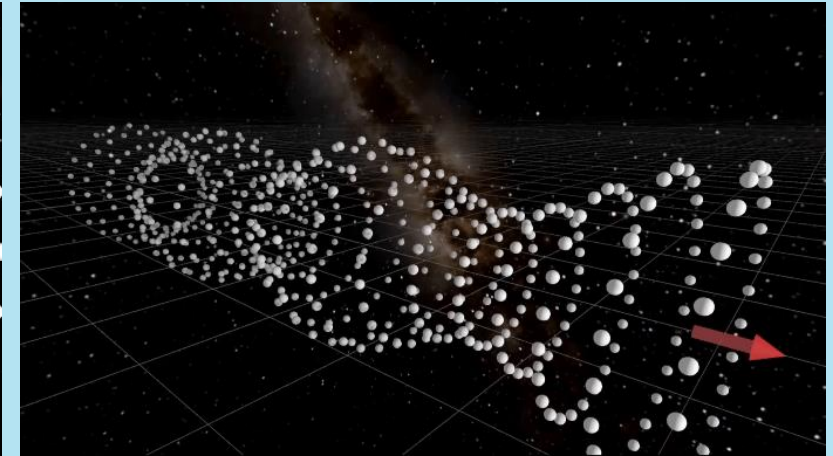
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GW out of screen

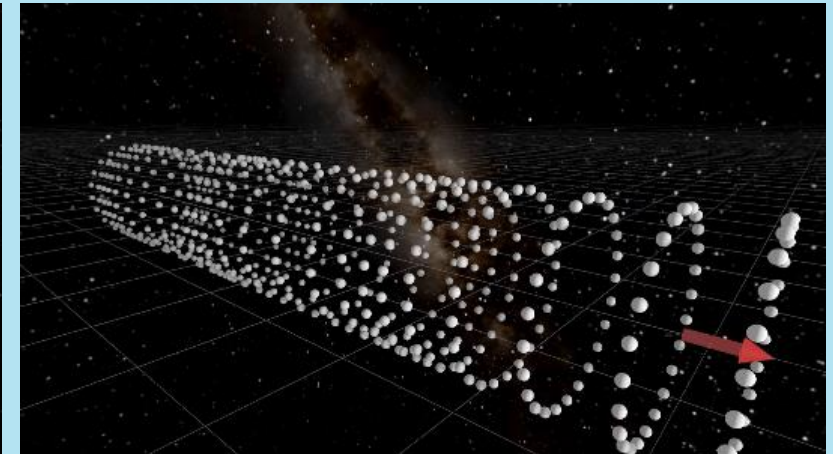
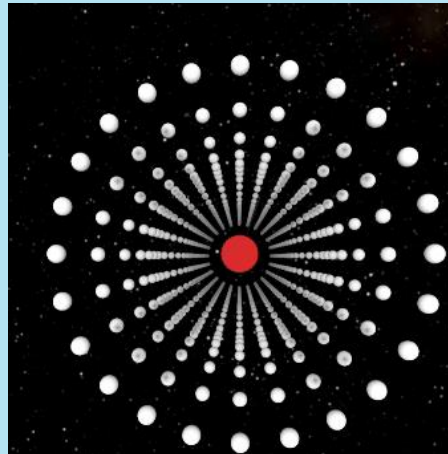


45° view, GW along direction of arrow



b

L

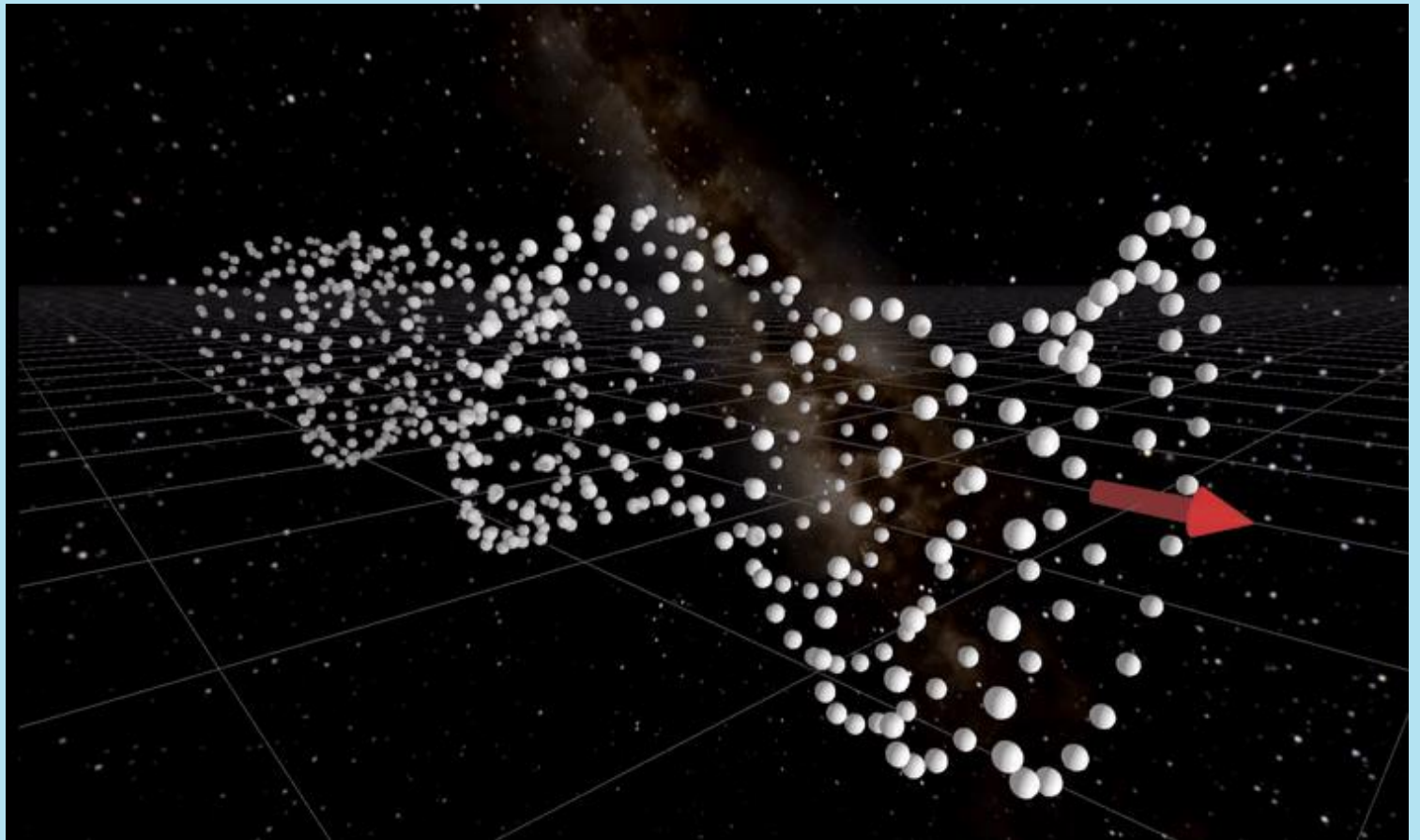


Beware: field content $\neq$ polarizations!

Theory	Field content	Strong EP	Massless graviton	Lorentz symmetry	Linear $T_{\mu\nu}$	Weak EP	Well-posed?	Weak-field constraints
Extra scalar field								
Scalar-tensor	S	X	✓	✓	✓	✓	✓ [34]	[35–37]
Multiscalar	S	X	✓	✓	✓	✓	✓ [38]	[39]
Metric $f(R)$	S	X	✓	✓	✓	✓	✓ [40, 41]	[42]
Quadratic gravity								
Gauss-Bonnet	S	X	✓	✓	✓	✓	✓?	[43]
Chern-Simons	P	X	✓	✓	✓	✓	X ✓? [44]	[45]
Generic	S/P	X	✓	✓	✓	✓	?	
Horndeski	S	X	✓	✓	✓	✓	✓?	
Lorentz-violating								
Æ-gravity	SV	X	✓	X	✓	✓	✓?	[46–49]
Khronometric/ Hořava-Lifshitz	S	X	✓	X	✓	✓	✓?	[48–51]
n-DBI	S	X	✓	X	✓	✓	?	none ([52])
Massive gravity								
dRGT/Bimetric	SVT	X	X	✓	✓	✓	?	[17]
Galileon	S	X	✓	✓	✓	✓	✓?	[17, 53]
Nondynamical fields								
Palatini $f(R)$	–	✓	✓	✓	X	✓	✓	none
Eddington-Born-Infeld	–	✓	✓	✓	X	✓	?	none
Others, not covered here								
TeVSe	SVT	X	✓	✓	✓	✓	?	[37]
$f(R)\mathcal{L}_m$	?	X	✓	✓	✓	X	?	
$f(T)$	?	X	✓	X	✓	✓	?	[54]

An example: Einstein-æther theory

- This theory allows for all six polarizations



Animation credit: [\[Schumacher et al. ISVC, 2024; POINT\]](#)

Why Einstein-æther theory?

- One way to modify gravity is to break Lorentz-invariance
 - Lorentz invariance = the outcome of experiments are independent of the reference frame
 - This theory is the most general Lorentz-violating theory (with single vector field and its derivatives, that still gives 2nd order field equations)
- Some attempts to unify gravity and quantum mechanics
⇒ Lorentz violation

The action in Einstein-æther Theory

Ricci Scalar in GR

$$\mathcal{L}_{\text{æ}} = R + \lambda(U^\mu U_\mu - 1) + \frac{1}{3}c_\theta \theta^2 + c_\sigma \sigma_{\mu\nu} \sigma^{\mu\nu} + c_\omega \omega_{\mu\nu} \omega^{\mu\nu} + c_a A_\mu A^\mu$$

æther field
4-velocity

expansion

shear

vorticity

acceleration

[Jacobson, Mattingly, PRD, 2001.]

**Constraining the coupling constants
constrains the theory**

Field Equations in this theory

To get to the field equations, vary the action with respect to the metric, the æther field and the Lagrange multiplier

$$E_{\alpha\beta} \equiv G_{\alpha\beta} - T_{\alpha\beta}^{\text{æ}} - 8\pi G T_{\alpha\beta}^{\text{mat}} = 0$$

$$\mathbb{E}_{\mu} \equiv \left[\nabla_{\alpha} J^{\alpha\nu} - \left(c_a - \frac{c_{\sigma} + c_{\omega}}{2} \right) A_{\alpha} \nabla^{\nu} U^{\alpha} \right] h_{\mu\nu} = 0.$$

$$J^{\alpha}_{\mu} \equiv M^{\alpha\beta}_{\mu\nu} \nabla_{\beta} U^{\nu}, \quad (2.9)$$

$$M^{\alpha\beta}_{\mu\nu} \equiv \left(\frac{c_{\sigma} + c_{\omega}}{2} \right) h^{\alpha\beta} g_{\mu\nu} + \left(\frac{c_{\theta} - c_{\sigma}}{3} \right) \delta_{\mu}^{\alpha} \delta_{\nu}^{\beta} + \left(\frac{c_{\sigma} - c_{\omega}}{2} \right) \delta_{\nu}^{\alpha} \delta_{\mu}^{\beta} + c_a U^{\alpha} U^{\beta} g_{\mu\nu}. \quad (2.10)$$

$$\begin{aligned} T_{\alpha\beta}^{\text{æ}} = & \nabla_{\mu} \left(J_{(\alpha}^{\mu} U_{\beta)} - J^{\mu}_{(\alpha} U_{\beta)} - J_{(\alpha\beta)} U^{\mu} \right) \\ & + \frac{c_{\omega} + c_{\sigma}}{2} [(\nabla_{\mu} U_{\alpha})(\nabla^{\mu} U_{\beta}) - (\nabla_{\alpha} U_{\mu})(\nabla_{\beta} U^{\mu})] \\ & + U_{\nu} (\nabla_{\mu} J^{\mu\nu}) U_{\alpha} U_{\beta} \\ & - \left(c_a - \frac{c_{\sigma} + c_{\omega}}{2} \right) [A^2 U_{\alpha} U_{\beta} - A_{\alpha} A_{\beta}] \\ & + \frac{1}{2} M^{\sigma\rho}_{\mu\nu} \nabla_{\sigma} U^{\mu} \nabla_{\rho} U^{\nu} g_{\alpha\beta}, \end{aligned} \quad (2.8)$$

Einstein-æther constraints

Motivation	Constraint	Reference(s)
Avoiding gradient instabilities/ghosts	$c_T^2 > 0, \quad c_S^2 > 0, \quad c_V^2 > 0$	[Jacobson, Mattingly, PRD, 2004.] [Garfinkle, Jacobson, PRL, 2011.]
Positive energy conditions	$0 \leq c_a \leq 2, \quad c_\omega \geq -\frac{c_\sigma}{1 - c_\sigma}$	[Foster, PRD, 2006.] [Eling, PRD, 2009.]
GW170817 & GRB170817A	$-3 \times 10^{-15} < c_T - 1 < 7 \times 10^{-16}$ $\Rightarrow c_\sigma \approx \mathcal{O}(10^{-15})$	[Abbott et al., Astrophys J. 2017.]
Big Bang Nucleosynthesis	$c_\theta \geq 0, \quad c_\theta + \frac{8c_a}{7} \leq \frac{2}{7}$	[Carroll, Lim, PRD. 2004.]
No gravitational Cherenkov radiation	$c_T \geq 1, \quad c_V \geq 1, \quad c_S \geq 1$	[Elliott, Moore, Stoica, JHEP, 2005.]
Solar system experiments	$ \alpha_1 \leq 10^{-4}$ $ \alpha_2 \leq 10^{-7}$	[Muller et al., Astrophys. Space Sci., 2008.] [Will, Living Rev. Rel., 2014.]
Binary pulsar studies	$\alpha_1 \sim (-0.563 \pm 1.021) \times 10^{-5}$	[Gupta et al., CQG, 2021.]

Propagation speeds – specific example

As functions of the coupling constants in Einstein-æther theory, the speeds of the different polarizations are:

$$c_T^2 = \frac{1}{1 - c_\sigma}$$

$$c_V^2 = \frac{c_\sigma + c_\omega - c_\sigma c_\omega}{2c_a(1 - c_\sigma)}$$

$$c_S^2 = \frac{(c_\theta + 2c_\sigma)(1 - c_a/2)}{3c_a(1 - c_\sigma)(1 + c_\theta/2)}$$

In the limit $c_\sigma \rightarrow 0$,

$$c_T^2 \rightarrow 1$$

$$c_V^2 = \frac{c_\omega}{2c_a}$$

$$c_S^2 = \frac{c_\theta(1 - c_a/2)}{3c_a(1 + c_\theta/2)}$$

Waves in Einstein-æther theory

$$\tilde{h}_{(+,2)}(f) = A_{(2)}(f) [(1 + \cos^2 \iota)] e^{i\Psi_{(2)}} e^{-i2\pi f D_L (1 - c_T^{-1})},$$

$$\tilde{h}_{(\times,2)}(f) = A_{(2)}(f) [2i \cos \iota] e^{i\Psi_{(2)}} e^{-i2\pi f D_L (1 - c_T^{-1})},$$

$$\tilde{h}_{(b,2)}(f) = A_{(2)}(f) \left[\frac{1}{2 - c_a} \left(3c_a(Z - 1) - \frac{2\mathcal{S}}{c_S^2} \right) \sin^2 \iota \right] e^{i\Psi_{(2)}} e^{-i2\pi f D_L (1 - c_S^{-1})},$$

$$\tilde{h}_{(L,2)}(f) = a_{bL} \tilde{h}_{(b,2)}(f),$$

$$\tilde{h}_{(X,2)}(f) = A_{(2)}(f) \left[\frac{\beta_1}{c_\sigma + c_\omega - c_\sigma c_\omega} \frac{1}{2c_V} \left(\mathcal{S} - \frac{c_\sigma}{1 - c_\sigma} \right) \sin(2\iota) \right] e^{i\Psi_{(2)}} e^{-i2\pi f D_L (1 - c_V^{-1})},$$

$$\tilde{h}_{(Y,2)}(f) = A_{(2)}(f) \left[\frac{i\beta_1}{c_\sigma + c_\omega - c_\sigma c_\omega} \frac{1}{c_V} \left(\mathcal{S} - \frac{c_\sigma}{1 - c_\sigma} \right) \sin(\iota) \right] e^{i\Psi_{(2)}} e^{-i2\pi f D_L (1 - c_V^{-1})},$$

$$\tilde{h}_{(b,1)}(f) = A_{(1)}(f) \left[\frac{2i}{(2 - c_a)c_S} \sin \iota \right] e^{i\Psi_{(1)}} e^{-i2\pi f D_L (1 - c_S^{-1})},$$

$$\tilde{h}_{(L,1)}(f) = a_{bL} \tilde{h}_{(b,1)}(f),$$

$$\tilde{h}_{(X,1)}(f) = A_{(1)}(f) \left[\frac{i\beta_1}{c_\sigma + c_\omega - c_\sigma c_\omega} \cos \iota \right] e^{i\Psi_{(1)}} e^{-i2\pi f D_L (1 - c_V^{-1})},$$

$$\tilde{h}_{(Y,1)}(f) = A_{(1)}(f) \left[-\frac{\beta_1}{c_\sigma + c_\omega - c_\sigma c_\omega} \right] e^{i\Psi_{(1)}} e^{-i2\pi f D_L (1 - c_V^{-1})},$$

$$\beta_1 = -\frac{2c_\sigma}{c_V},$$

ppE framework extended

- Original paper (2009), extensions to all polarizations (2012)

$$\tilde{h} = \tilde{h}_{GR} e^{i\beta_{ppE} u_2^b} + \left[\sum_N \alpha_N F_N \right] u_2^a e^{-i\psi_{GR}} e^{i\beta_{ppE} u_2^b} + (\ell = 1 \text{ terms})$$

where $N \in \{+, \times, b, L, X, Y\}$

and polarizations with different propagation speeds (2023)

$$\tilde{h} = \tilde{h}_{GR} e^{i\beta_{ppE} u_2^b} + \left[\sum_N \alpha_N F_N e^{-2i\pi f D(1-v_N^{-1})} \right] u_2^a e^{-i\psi_{GR}} e^{i\beta_{ppE} u_2^b} + (\ell = 1 \text{ terms})$$

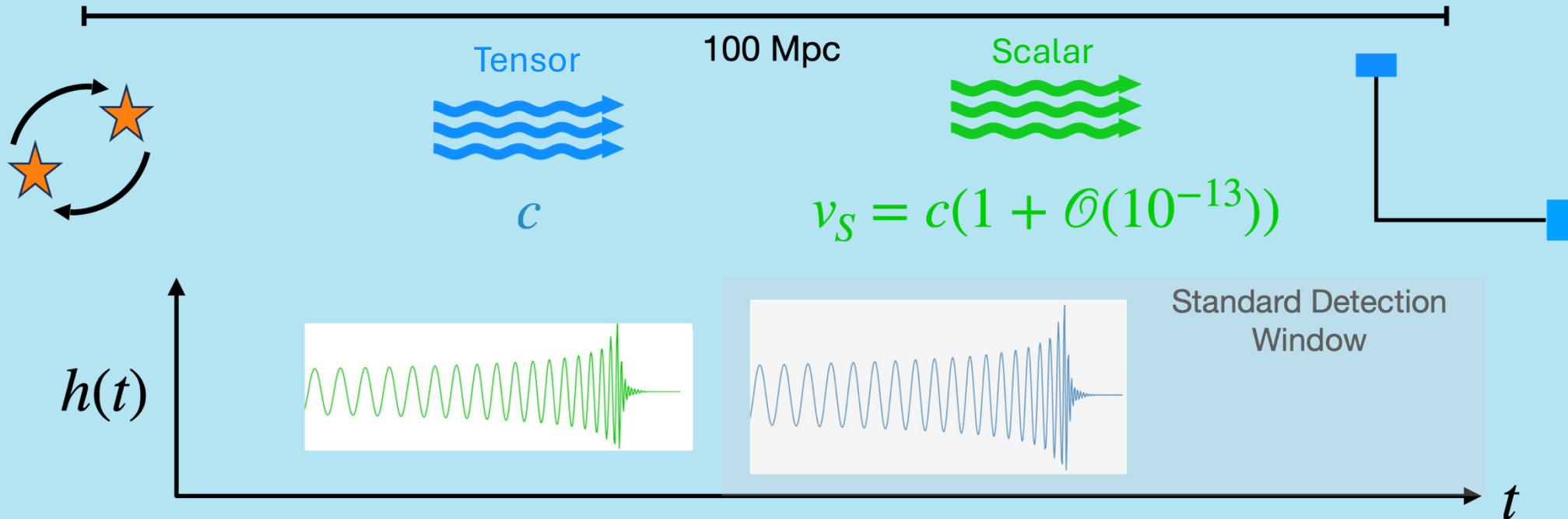
Propagation speed constraints

- If $v_{S,V,T} < c$, then massive particles can travel faster than the GW $\rightarrow$ producing a “gravitational Cherenkov type radiation”
- However, high energy cosmic rays rule out this Cherenkov radiation
 $\rightarrow v_{S,V,T} \geq c$ [Moore and Nelson, JHEP, 2001] [Kiyota & Yamamoto, PRD, 2015]
- GW170817 & GRB170817A give $v_T = c \pm \mathcal{O}(10^{-15})$ [Abbott et al., ApJ, 2017]

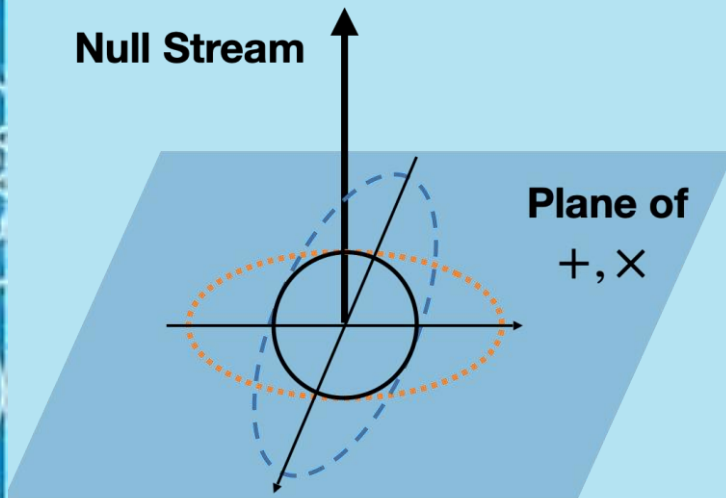
Additional modes must arrive **with** or **before** the tensor modes.

Different arrival times for chirping signals

- Even a tiny difference in speed can build up over large distances
- E.g. [Schumacher, Yagi, and Yunes, PRD, 2023]



Null streams



- A way to separate signals from noise in a detector network by constructing linear combinations of data that contain no signal [Guersel, Tinto, PRD, 1989]
 - Extended by [Chatziioannou, Yunes, Cornish, PRD, 2012]
- Project data to direction orthogonal to that defined by angular pattern function
- Any detection within the null stream would indicate a deviation from GR [Chatziioannou, Yunes, Cornish, PRD, 2012]

Null streams – a numbers game

- The number of null streams depends on the number of detectors with *different* orientations
- Need 3 detectors to be able to construct 1 GR null stream [[Chatziioannou, Yunes, Cornish, PRD, 2012](#)]
- With more than 5 detectors one could construct complete null streams in any theory of gravity [[Chatziioannou, Yunes, Cornish, PRD, 2012](#)]

Current detectors = 4

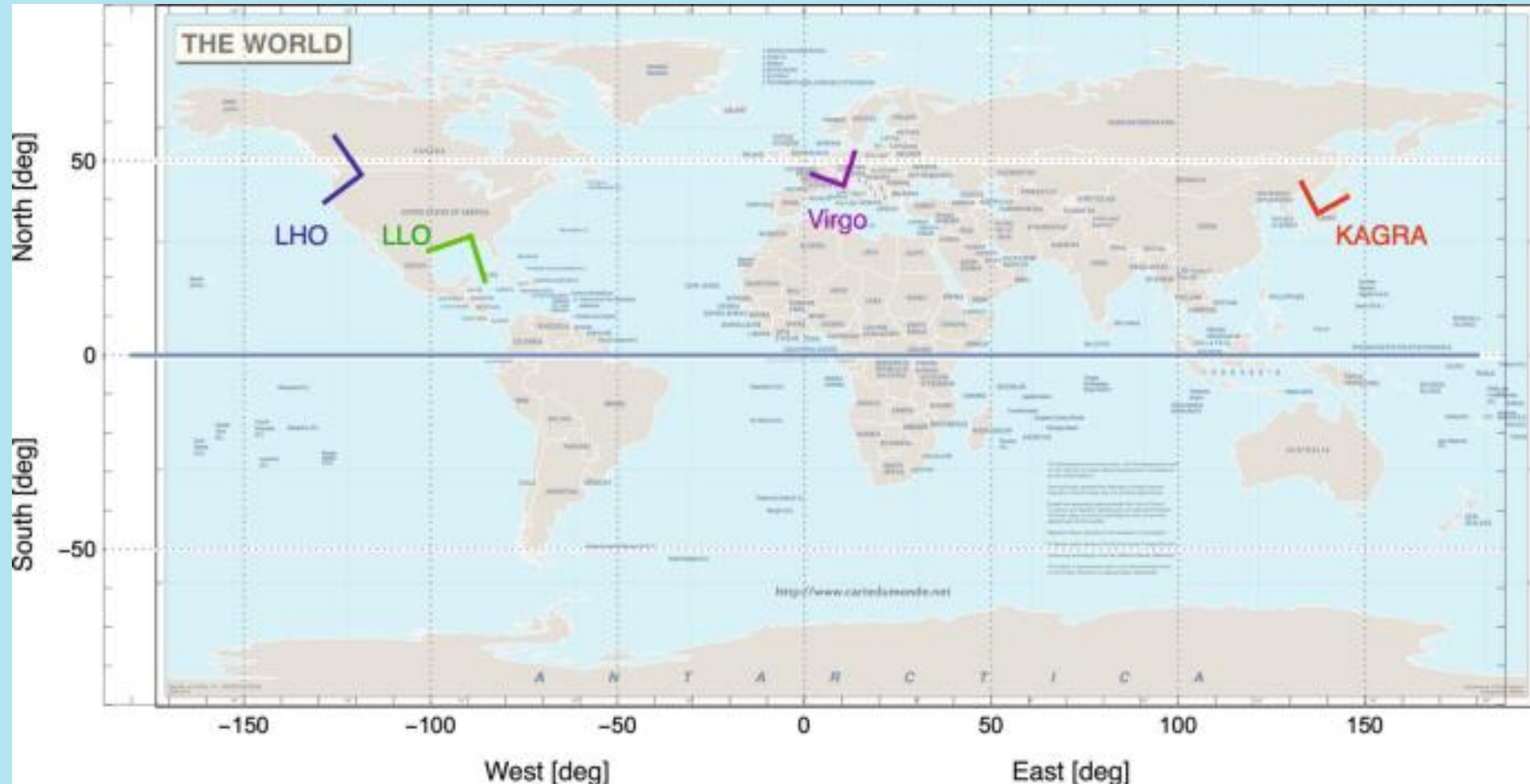


Image Credit:
[Handbook of
GW
astronomy,
2020]

Current detectors = ~~4~~ = 3

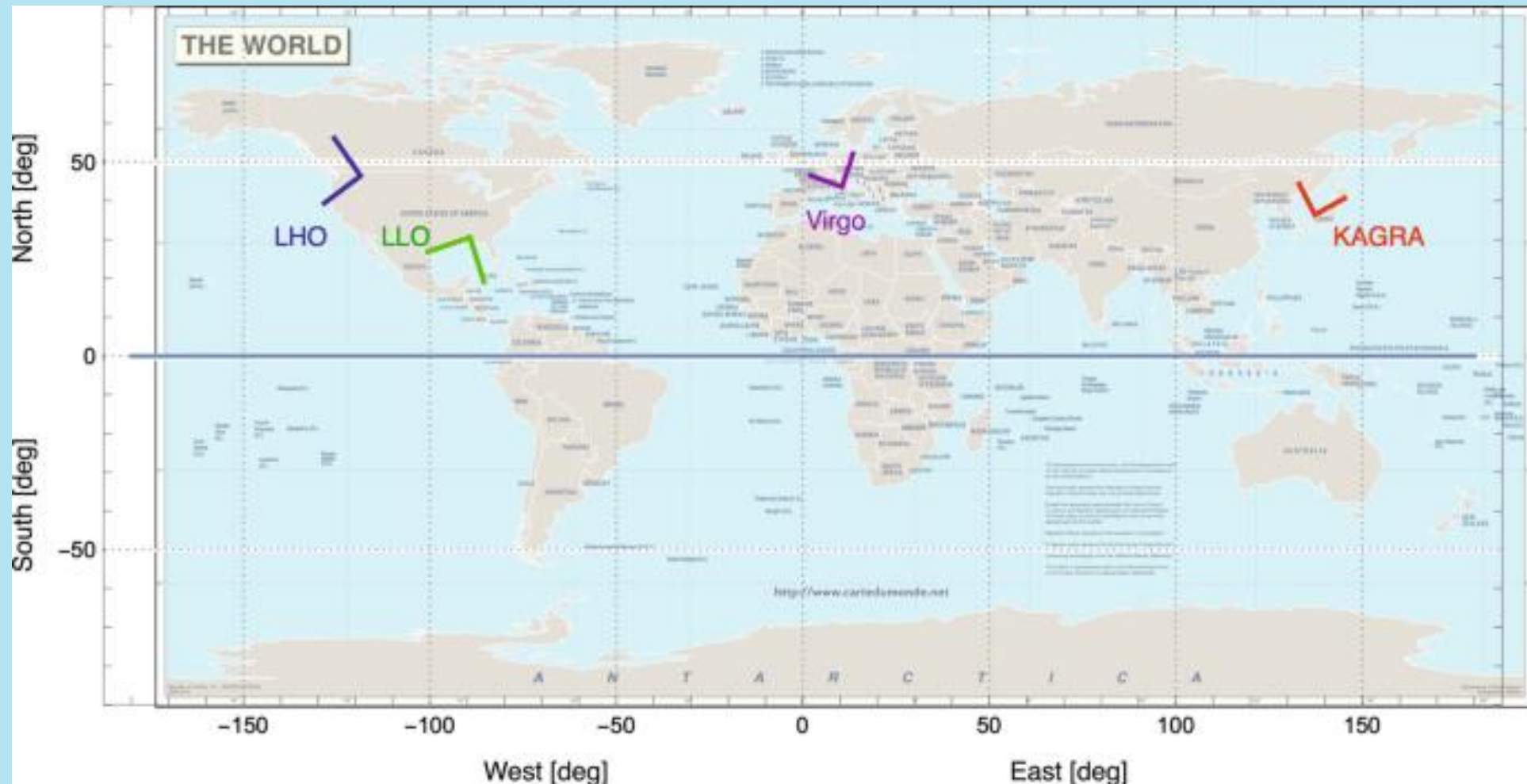


Image Credit:
[Handbook of
GW
astronomy,
2020]

Current detectors = ~~4~~ = ~~3~~ = 2

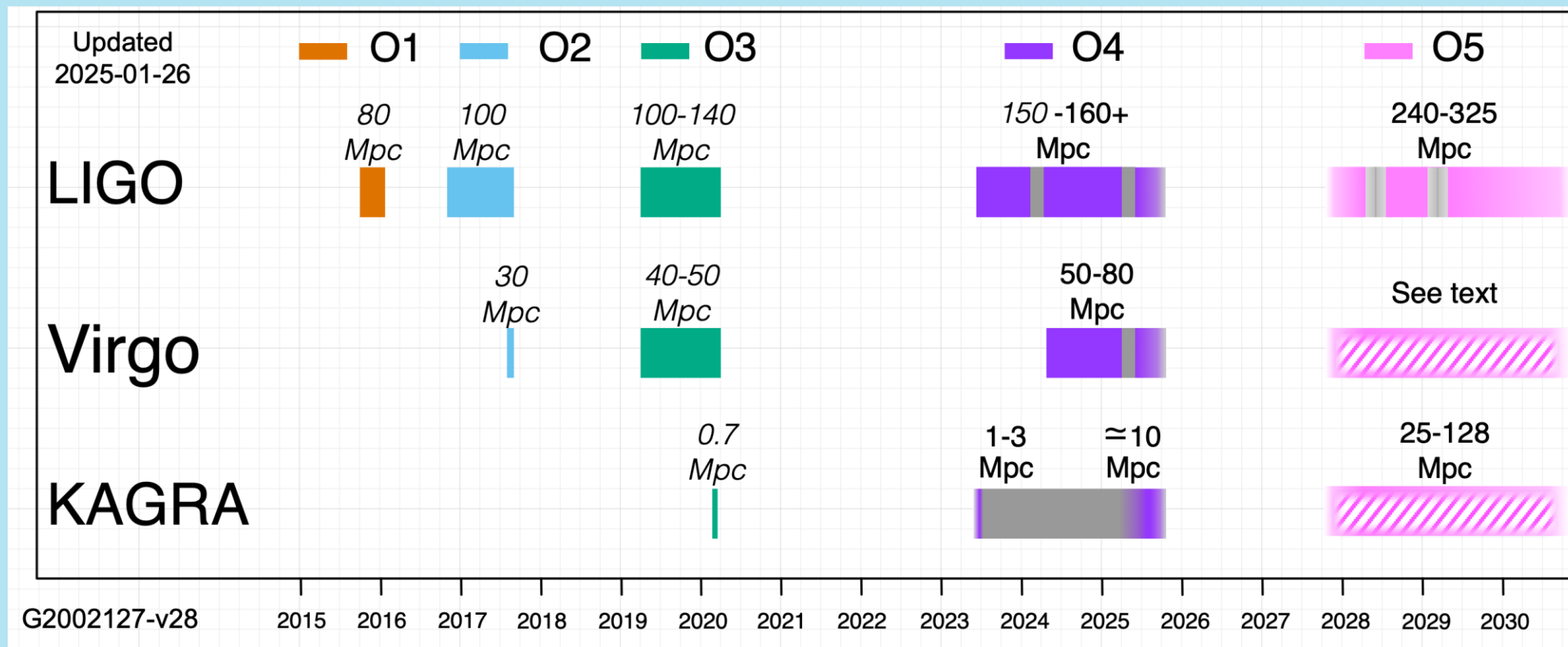


Image Credit: LVK Observing Plans, Jan. 25, 2025

Current detectors = ~~4~~ = ~~3~~ $\approx$ 2

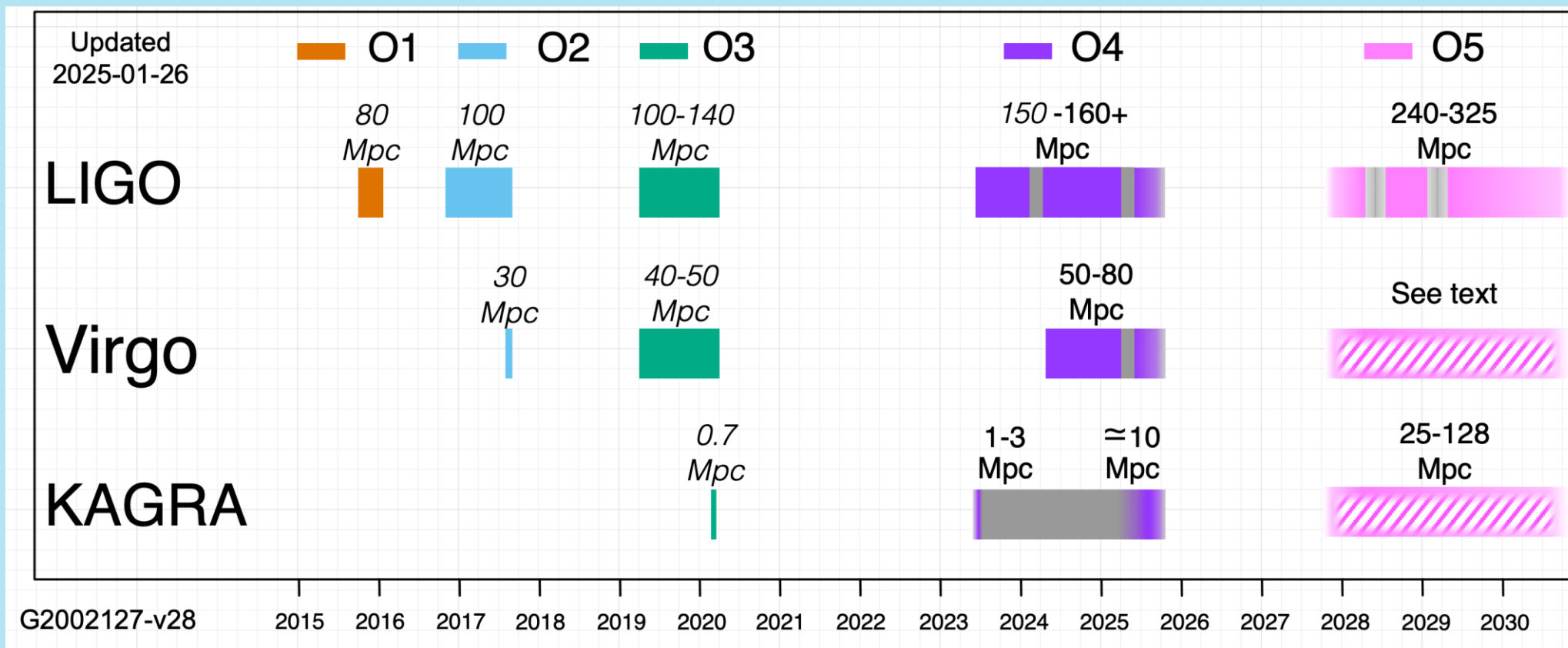
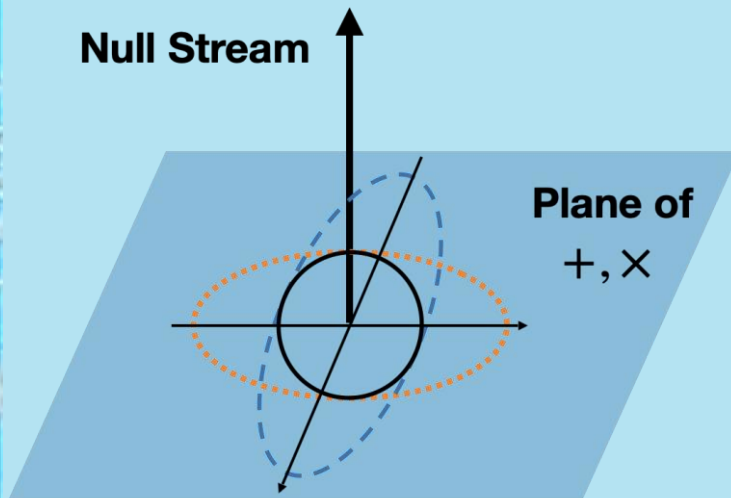


Image Credit: LVK Observing Plans, Jan. 25, 2025

Another problem with null streams



- **Beware:** absence of additional polarizations in null stream only constrains polarizations that travel at the same speed!
 - But it is possible that $v_T \neq v_V \neq v_S$
 - The null channel may be empty because additional polarizations arrived at a different time

A decorative vertical bar on the left side of the slide, featuring a blue background with white line-art icons of scientific equipment like microscopes, light bulbs, and molecular structures.

Additional polarizations not yet ruled out...

...PTAs are uniquely positioned to do something about this